MILITARY STANDARD

MIL-STD-1325-177
(NAVY)

RAILCAR LOADING OF HAZARDOUS MATERIALS

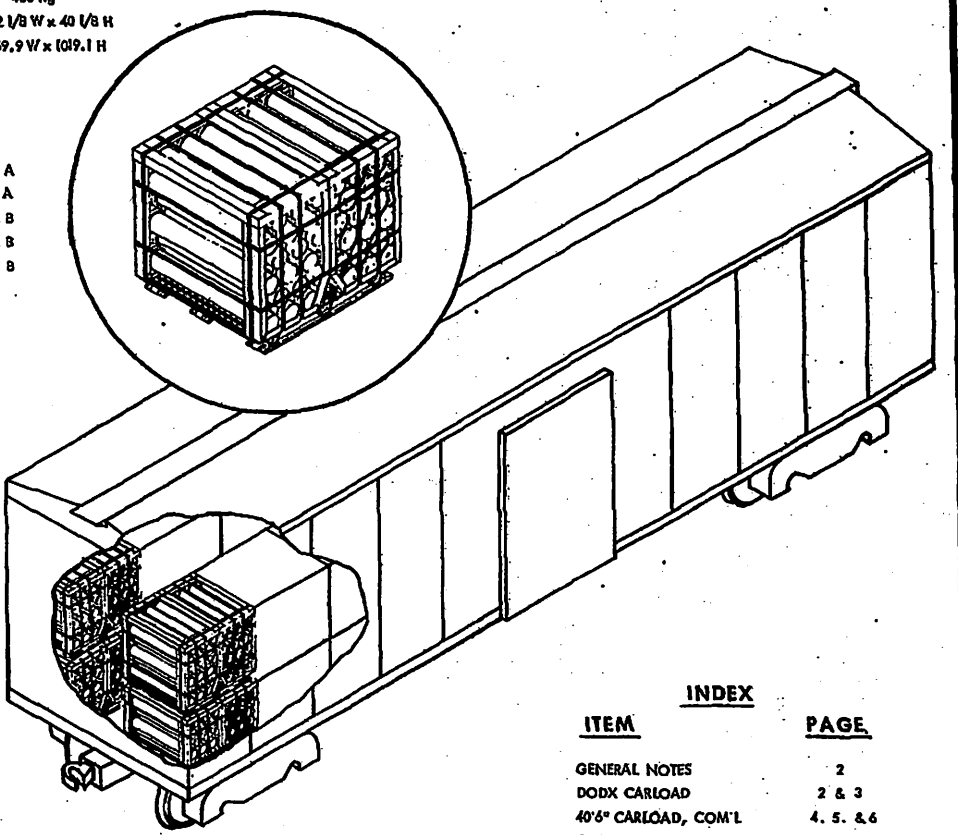
3 JUNE 1975

CARTRIDGE, 76MM/62 CAL OR CHARGE, CLEARING, 76MM/62 CAL
IN PALLET ADAPTER MK 121 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING 54/263
WEIGHT (APPROX)
CARTRIDGE ..... 1511 LBS
685 Kg
CLEARING CHARGE .....949 LBS
430 Kg
DIMENSIONS (INCHES)...48 3/4 L x 42 1/8 W x 40 1/8 H
(MM).....1238.3L x 1069.9 W x 1019.1 H
CUBE (CU FT) 44.7
(CU METER) 1.2657
DOT HAZARD CLASSIFICATION

CARTRIDGE TYPE	CLASS
HE-IR	EXPLOSIVE A
HE-FD	EXPLOSIVE A
BL&P	EXPLOSIVE B
TPSP	EXPLOSIVE B
CLEARING	EXPLOSIVE B



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GENERAL NOTES	2
DODX CARLOAD	2 & 3
40'6" CARLOAD, COM'L	4, 5, & 6
50'6" CARLOAD, COM L	8, 9, & 10

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK -236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE

H S Seay NWHL 4/29/75
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J E Kell 4/30/75
SIGNATURE (SEA) SYS COM. BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

L. Y. Jaasmack 3/25/75
SIGNATURE SUPERVISOR, MILITARY & INTERMODEL SERVICES DATE

ORIGINATOR L R Causey 4/28/75
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

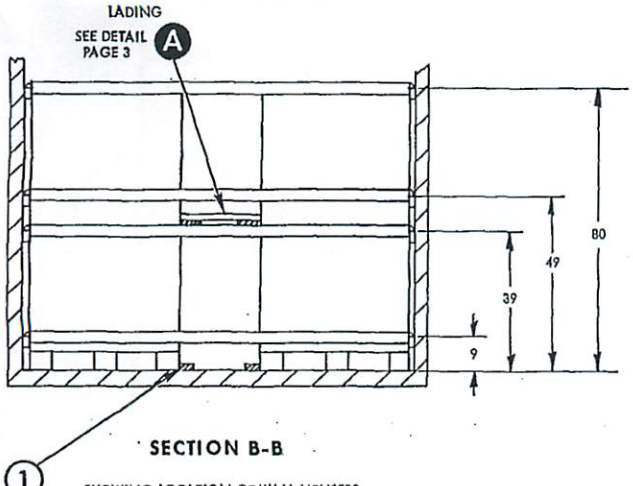
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT. 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAIL (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT. 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT. 6 INCHES INSIDE LENGTH. 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 76 MM/62 CAL CARTRIDGE OR 76 MM/62 CAL CLEARING CHARGE IN PALLET ADAPTER MK 121 MOD 0 PALLETIZED IN ACCORDANCE WITH WR-54/263.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP. RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 OF 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC MM-L-751
    NAILS - FED. SPEC FF-N-105
    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).
    SEALS - FED. SPEC. QQ-S-781, STYLE III, HEAVY DUTY.
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA. CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I)

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 44 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING ANY MIXTURE OF 76 MM/62 CAL CARTRIDGES OR 76 MM/62 CAL CLEARING CHARGES MAY BE LOADED IN THE CARLOAD. WHEN LOADING CLEARING CHARGES ADJUST THE CARLOAD WEIGHT ACCORDINGLY.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING



SECTION B-B

SHOWING LOCATION OF WALL MEMBERS AND DOORWAY MEMBERS USED FOR CROSSMEMBERS

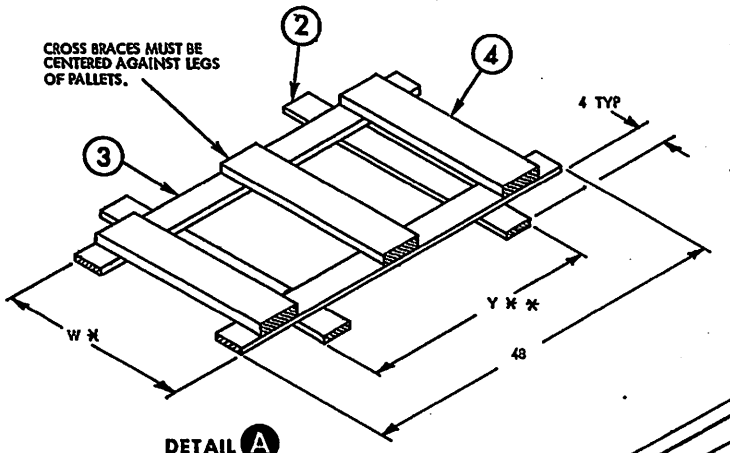
* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
4	CROSS BRACE	2 x 6 x W *	33	SEE 3	—	—
3	SWAY BRACE LONGITUDINAL	1 x 4 x 48	22	2, 4	SEE NOTE	**
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8) *	22	SEE 3	—	—
1	SLEEPER	2 x 4 x 48	22	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

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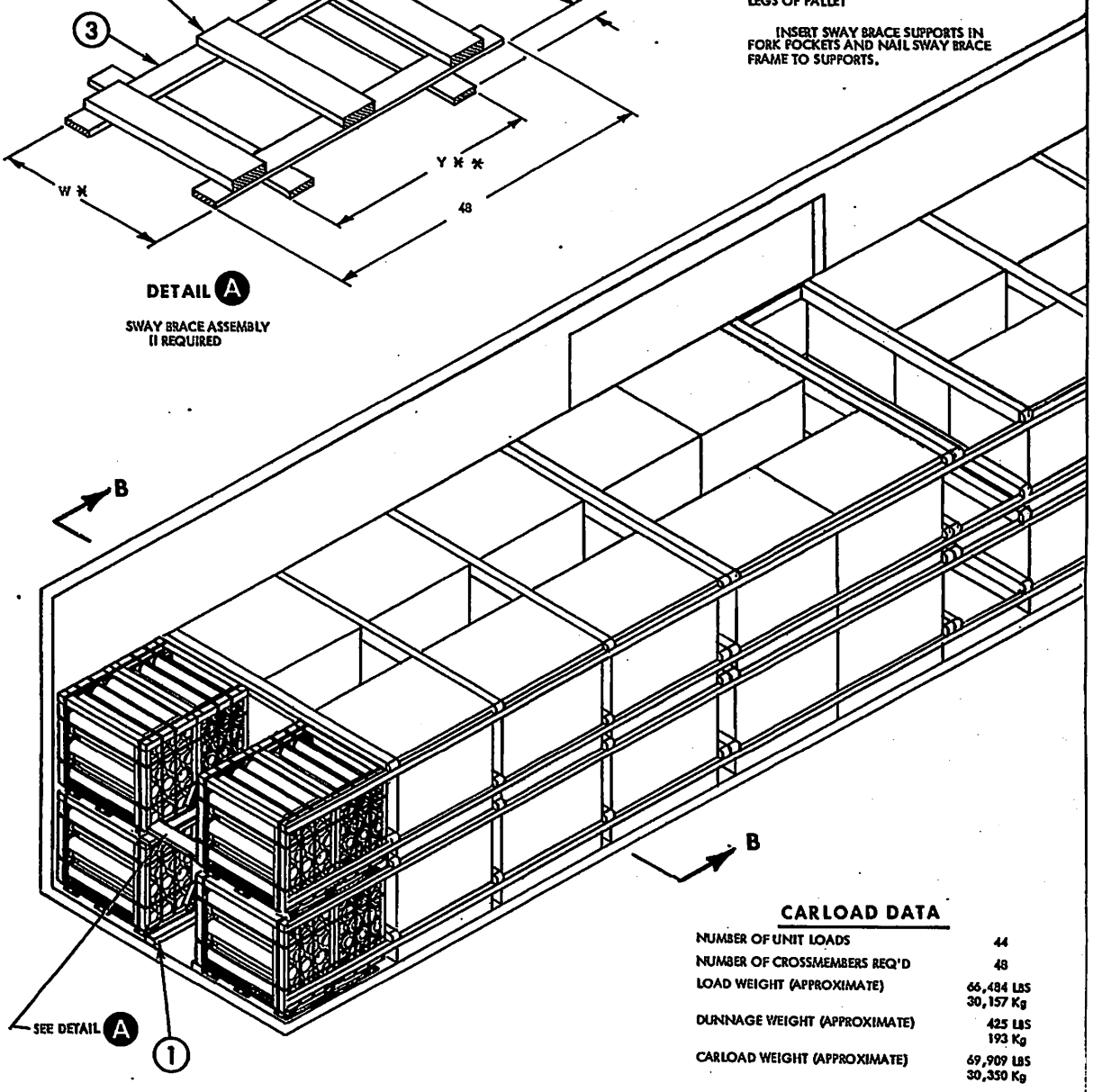
DETAIL A

SWAY BRACE ASSEMBLY (1 REQUIRED)

* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

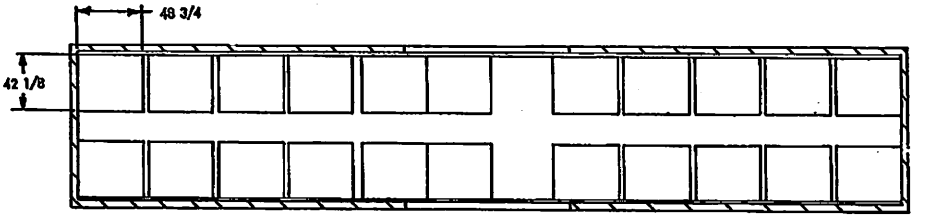
** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



CARLOAD DATA

NUMBER OF UNIT LOADS	44
NUMBER OF CROSSMEMBERS REQ'D	48
LOAD WEIGHT (APPROXIMATE)	66,484 LBS 30,157 Kg
DUNNAGE WEIGHT (APPROXIMATE)	425 LBS 193 Kg
CARLOAD WEIGHT (APPROXIMATE)	69,909 LBS 30,350 Kg

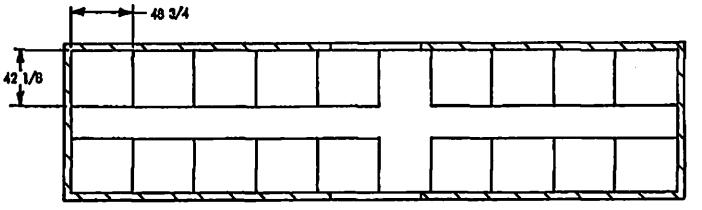


LOADING PLAIN

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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 36 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. ANY MIXTURE OF 76 MM/62 CAL CARTRIDGES OR 76 MM/62 CAL CLEARING CHARGES MAY BE LOADED IN THE CARLOAD. WHEN LOADING CLEARING CHARGES. ADJUST THE CARLOAD WEIGHT ACCORDINGLY.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT. WIDE.



LOADING PLAIN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS. REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
13	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 x 3 x 96	4	DOOR POST	2 PER FOOT	20d
11	STRUT	4 x 4 x WEDGE FIT***	16	6	2 PER JOINT	16d
10	HOLDDOWN CLEAT	2 x 3 x 36	4	9	5	16d
9	HOLDDOWN CLEAT	2 x 3 x 36	4	8	5	10d
8	HOLDDOWN SPACER	2 x 4 x CAR WIDTH-1	2	6	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 84	8	SEE 9	—	—
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W**	27	SEE 3	—	—
3	SWAY BRACE LONGITUDINAL	1 x 4 x 48	18	2 4	SEE NOTE	****
2	SWAY BRACE SUPPORT	1 x 4 x (W+6)**	18	SEE 3	—	—
1	SLEEPER	2 x 4 x 48*	18	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

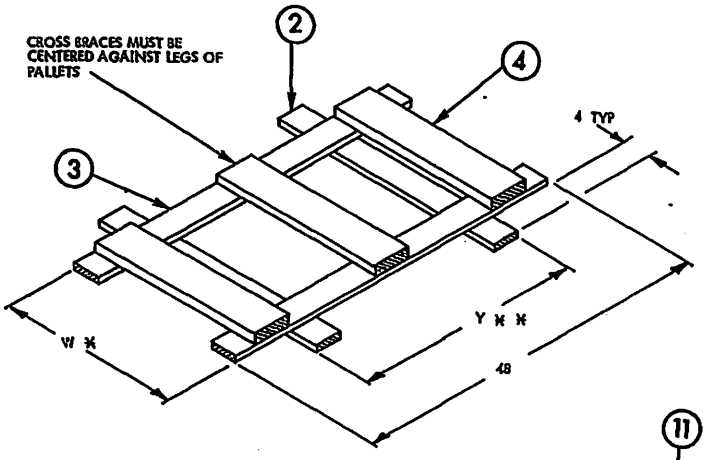
* USE SWAY BRACE ASSEMBLY DETAIL A FOR STEEL FLOOR CARS

** W= DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

*** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

**** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S

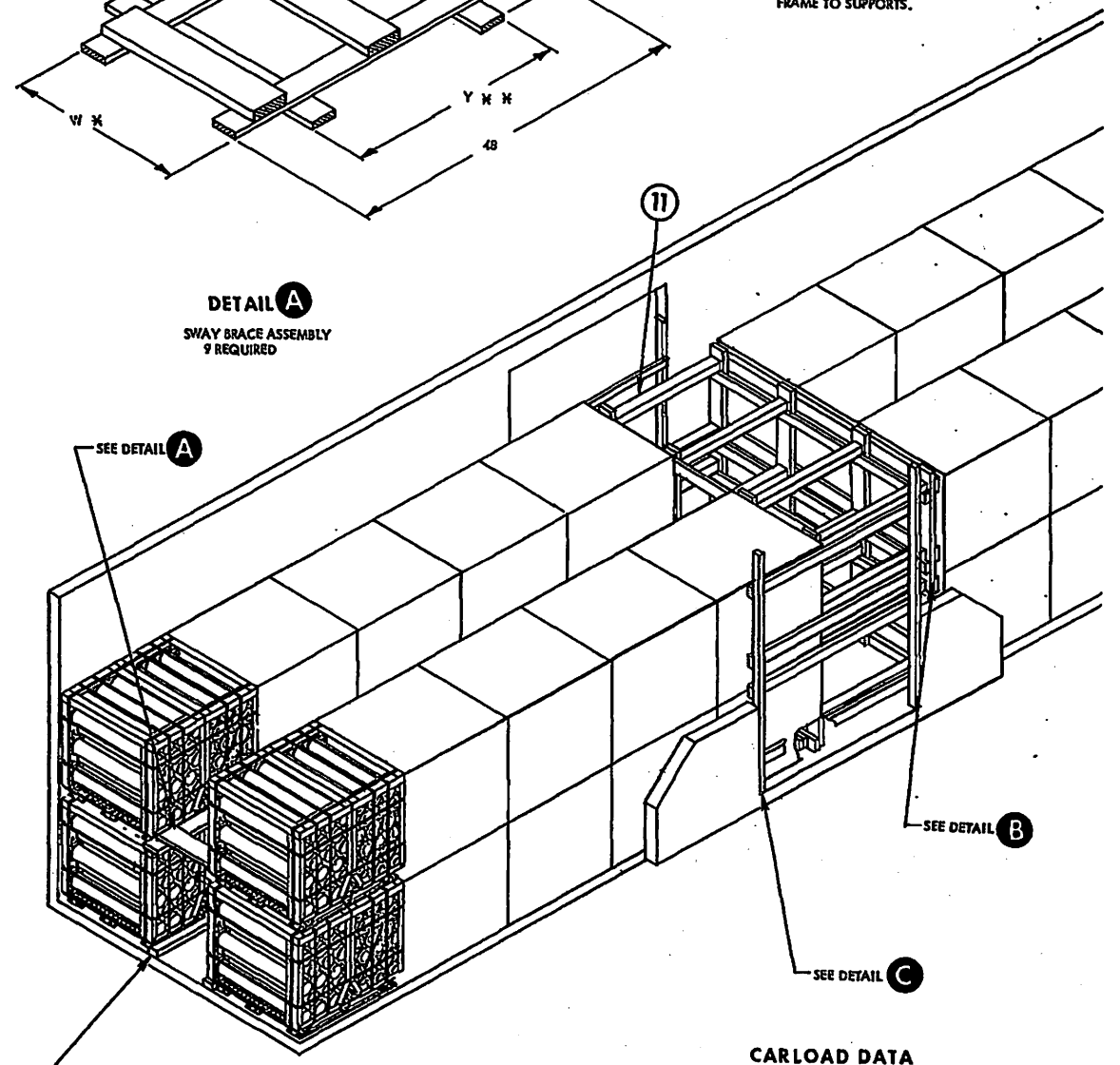
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* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



DETAIL A

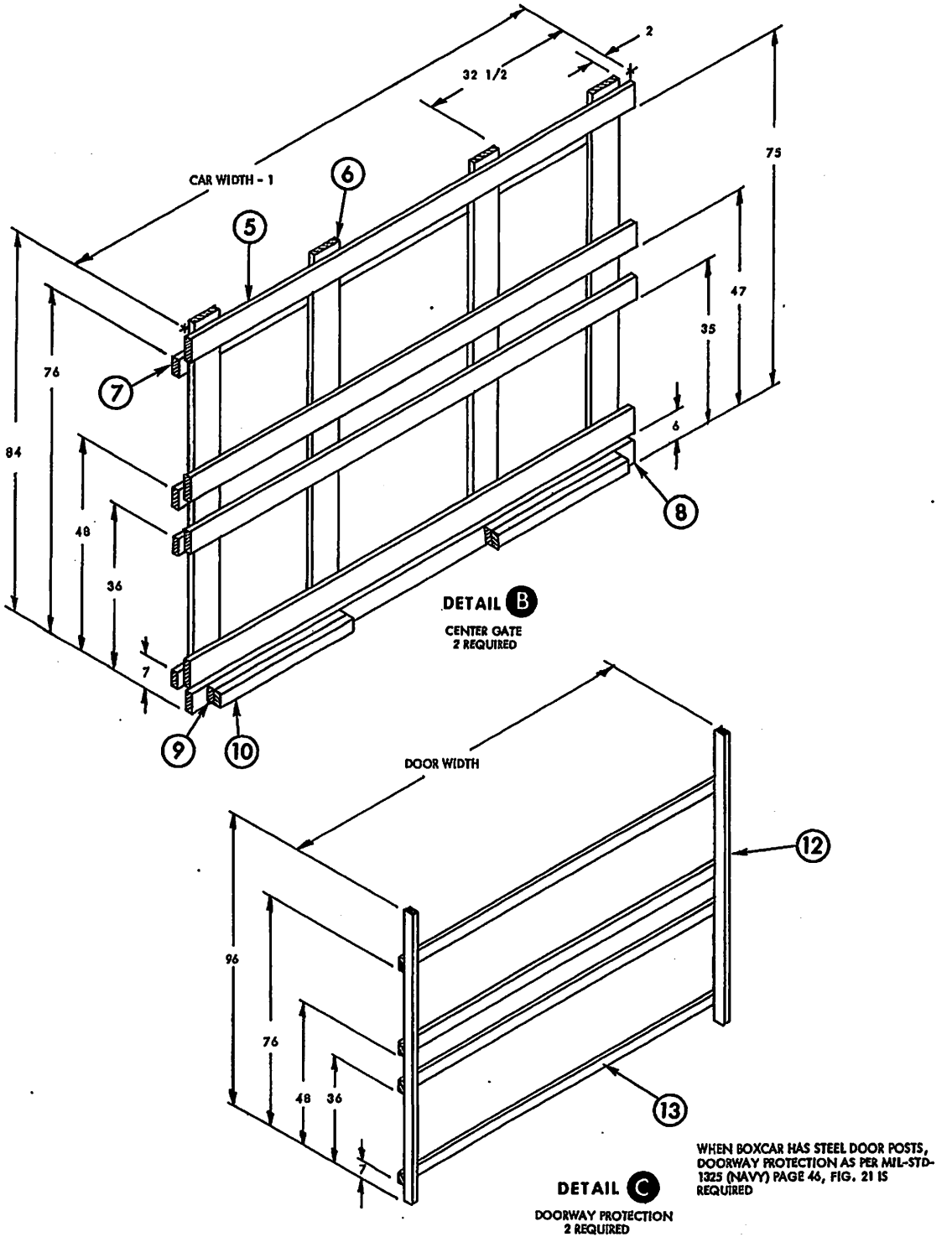
SWAY BRACE ASSEMBLY
9 REQUIRED

1 USE SWAY BRACE ASSEMBLY DETAIL A FOR STEEL FLOOR CARS

CARLOAD DATA

NUMBER OF UNIT LOADS	36
LOAD WEIGHT (APPROXIMATE)	54,396 LBS 24,674 Kg
DUNNAGE WEIGHT (APPROXIMATE)	1,135 LBS 515 Kg
CARLOAD WEIGHT (APPROXIMATE)	55,531 LBS 25,189 Kg

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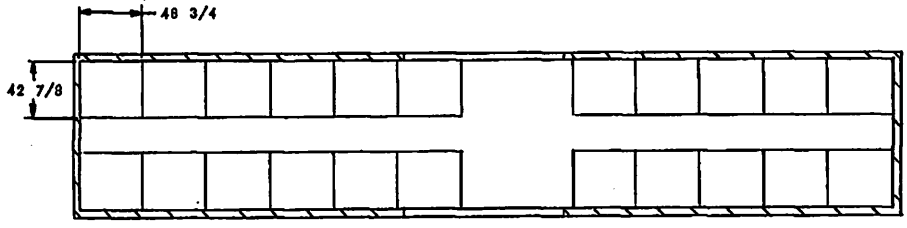
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 44 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. ANY MIXTURE OF 76MM/62 CAL CARTRIDGES OR 76MM/62 CAL CLEARING CHARGES MAY BE LOADED IN THE CARLOAD. WHEN LOADING CLEARING CHARGES ADJUST THE CARLOAD WEIGHT ACCORDINGLY.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR. WHETHER MAIN OR AUXILIARY EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR. AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREAS MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL D), MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. SECURE EACH STRAP TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DIMENSIONAL LUMBER DOORWAY PROTECTION PIECES 12 AND 13 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAIN

* USE SWAY BRACE ASSEMBLY DETAIL A FOR STEEL FLOOR CARS

** W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

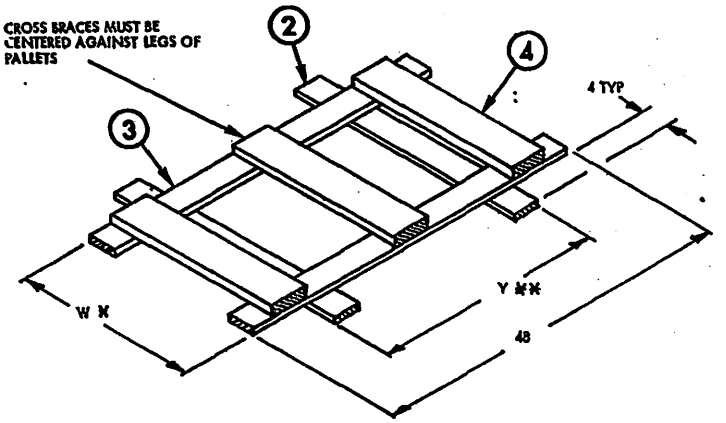
*** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

**** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S

PIECE NO.	DESCRIPTION	SIZE	NO PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE NAILS
13	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	12	2 PER JOINT	10d
12	VERTICAL DOORWAY MEMBER	2 x 3 x 96	4	DOOR POST	2 PER FOOT	20d
11	STRUT	4 x 4 x **** WEDGE FIT	16	6	2 PER JOINT	16d
10	HOLDDOWN CLEAT	2 x 3 x 36	4	9	5	16d
9	HOLDDOWN CLEAT	2 x 3 x 36	4	8	5	10d
8	HOLDDOWN SPACER	2x4 x CAR WIDTH-1	2	6	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 84	8	SEE 5	—	—
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	6	3 PER JOINT	10d
4	CROSSBRACE	2 x 6 x W **	33	SEE 3	—	—
3	SWAY BRACE LONGITUDINAL	1 x 4 x 48	22	2,4	SEE NOTE ***	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8) **	22	SEE 3	—	—
1	SLEEPER	2 x 4 x 48 *	22	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

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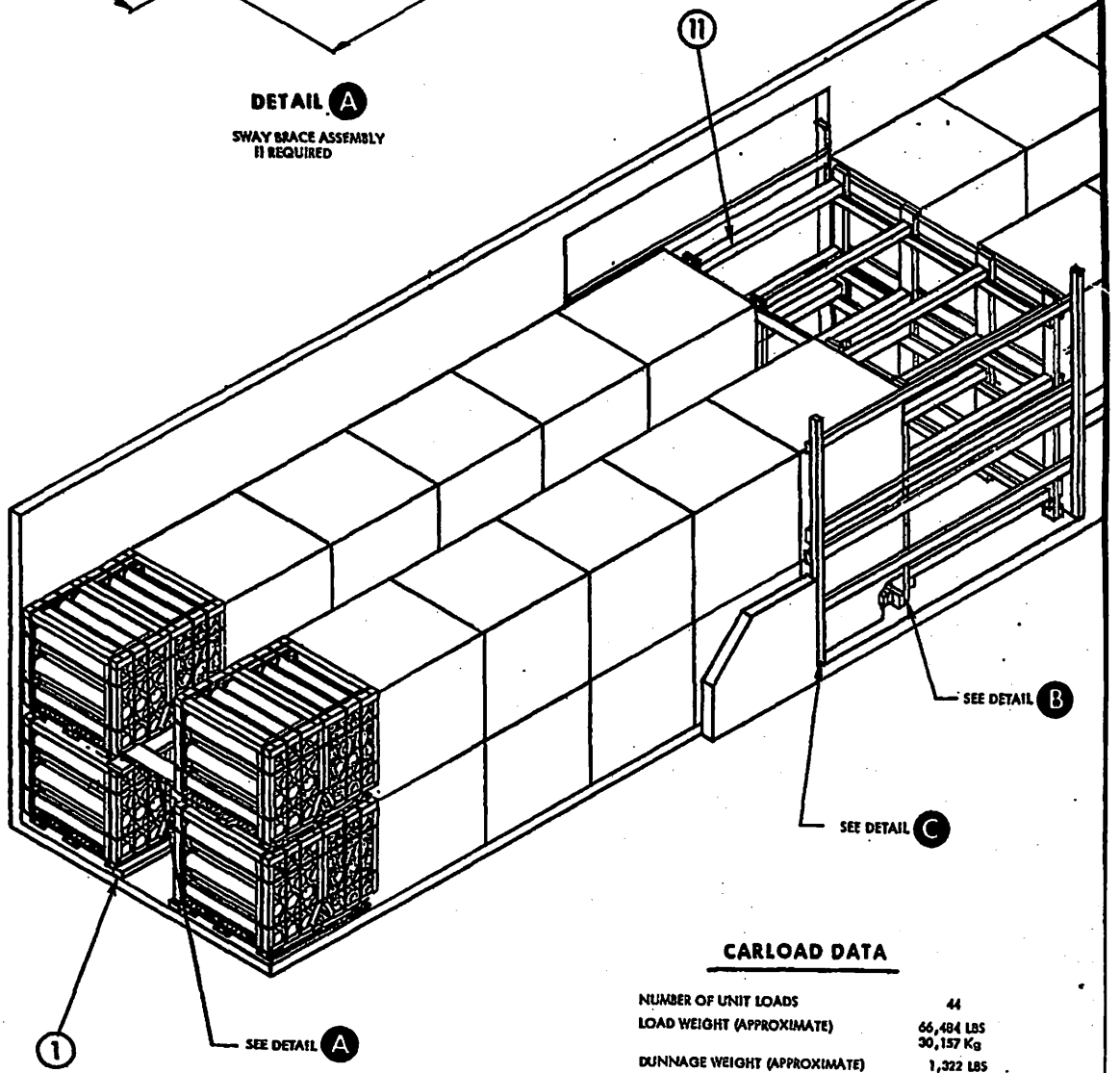
* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR

** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

SWAY BRACE ASSEMBLY IF REQUIRED

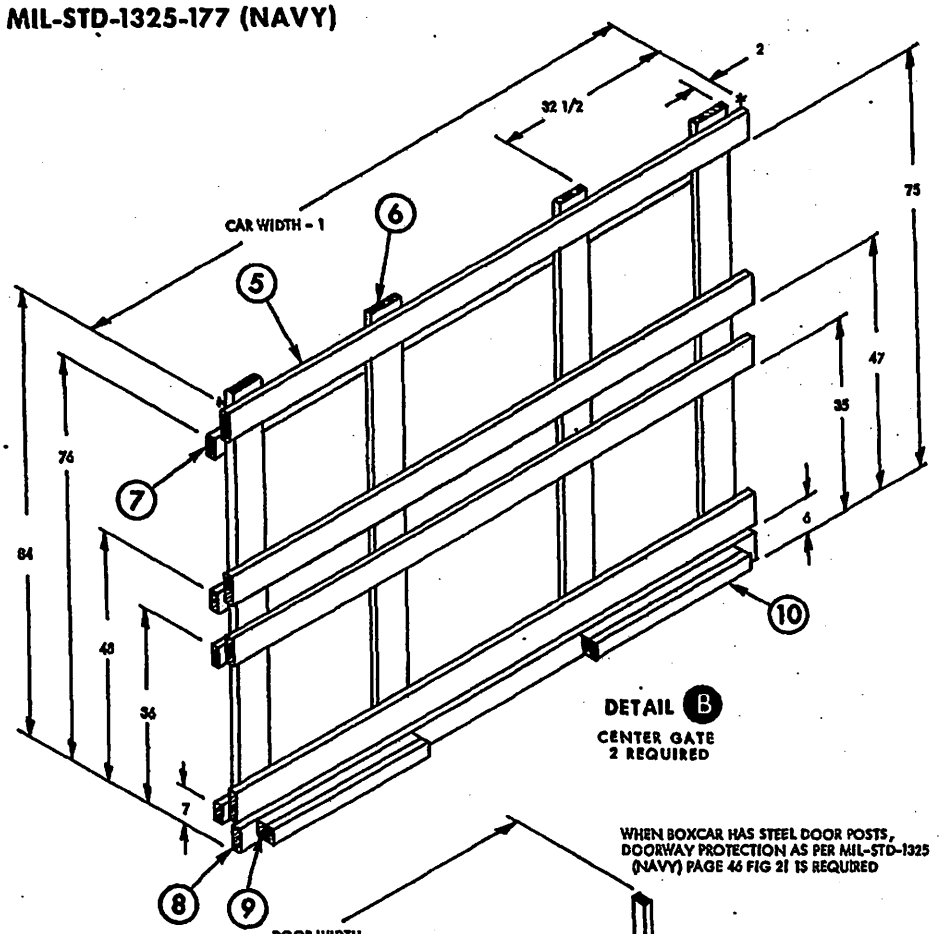


USE SWAY BRACE ASSEMBLY DETAIL A FOR STEEL FLOOR CARS

CARLOAD DATA

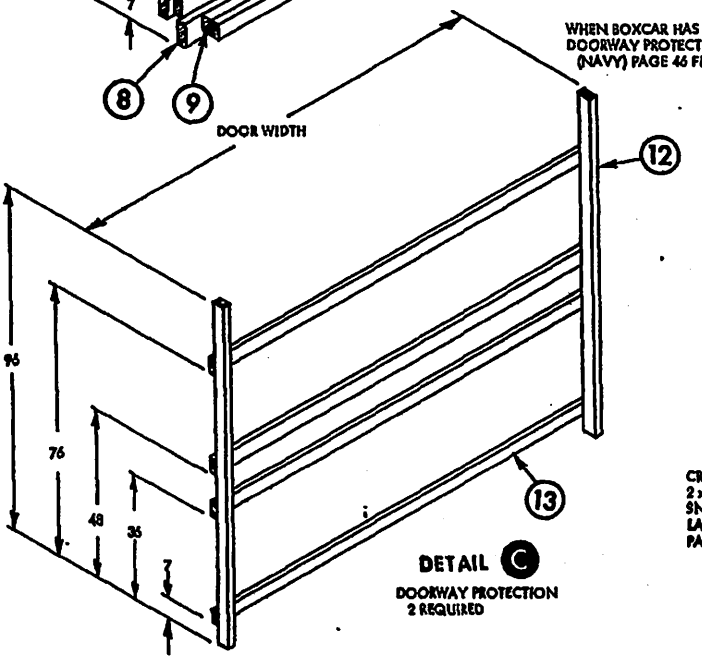
NUMBER OF UNIT LOADS	44
LOAD WEIGHT (APPROXIMATE)	66,484 LBS 30,157 Kg
DUNNAGE WEIGHT (APPROXIMATE)	1,322 LBS 600 Kg
CARLOAD WEIGHT (APPROXIMATE)	67,806 LBS 30,757 Kg

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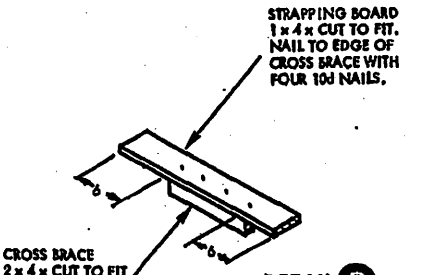


DETAIL B
CENTER GATE
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46 FIG 21 IS REQUIRED



DETAIL C
DOORWAY PROTECTION
2 REQUIRED



DETAIL D
TOP SWAY BRACE

REVIEW ACTIVITY
NAVY - OS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 8140- N 284)

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